

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Date: 30-08-2026

1. TITLE

Webpage Styling with External CSS3, Linear Gradients, Navigation Bar & Flexbox

2. AIM

To structure and style a multi-card webpage using external CSS3 stylesheets, linear gradients, responsive navigation bars, flexbox card grid layout, box shadows, and button hover transitions.

3. OBJECTIVE

The objectives of this experiment are:

- To link external CSS3 stylesheets to HTML5 documents.
 - To apply multi-color linear gradients and text shadows on headers.
 - To style a dark navigation bar with interactive hover color changes.
 - To implement responsive flexbox card layout with borders and shadows.
 - To create button hover color transitions.
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4. THEORY

External CSS separates document structure from visual presentation. CSS3 provides linear gradients, Flexbox for flexible horizontal/vertical alignment (display: flex, justify-content, gap), box shadows for modern UI depth, and pseudo-classes (:hover) for interactive transitions.

Application Flow:

HTML5 Structure -> External CSS Linking -> Gradient Header & Nav -> Flexbox Container -> Card Styling & Hover Effects

5. ALGORITHM / PROCEDURE

1. Create index.html and link external style.css using <link rel='stylesheet'>.
 2. Style header with linear-gradient(to right, #0066cc, #8000ff) and text shadow.
 3. Style navigation bar with dark background and yellow link hover color.
 4. Implement .container with display: flex, justify-content: center, and gap: 20px.
 5. Style .card with white background, rounded border, padding, and box shadow.
 6. Style buttons with green background and red hover color transition.
 7. Style footer with dark background.
 8. Verify styling and test negative cases (missing stylesheet, invalid colors).
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6. PROGRAM / SOURCE CODE

Source File: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
  <h1>My Styled Webpage</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Services</a>
  <a href="#">Contact</a>
</nav>

<section class="container">

  <div class="card">
    <h2>HTML5</h2>
    <p>HTML5 is used to create the structure of web pages.</p>
    <button>Read More</button>
  </div>

  <div class="card">
    <h2>CSS3</h2>
    <p>CSS3 is used to style web pages and improve appearance.</p>
    <button>Read More</button>
  </div>

  <div class="card">
    <h2>Web Design</h2>
    <p>HTML and CSS together create beautiful web applications.</p>
    <button>Read More</button>
  </div>

</section>

<footer>
  <p>© 2026 My Styled Webpage</p>
</footer>

</body>
</html>
```

Source File: style.css

```
body{
  margin:0;
  font-family:Arial,sans-serif;
  background:#f2f2f2;
}

header{
  background:linear-gradient(to right,#0066cc,#8000ff);
  color:white;
  text-align:center;
  padding:20px;
  text-shadow:2px 2px 5px gray;
}

nav{
  background:#333;
  text-align:center;
  padding:15px;
}

nav a{
  color:white;
  text-decoration:none;
  margin:20px;
  font-size:18px;
}

nav a:hover{
  color:yellow;
}

.container{
  display:flex;
  justify-content:center;
  gap:20px;
  margin:30px;
}

.card{
  background:white;
  width:250px;
  padding:20px;
  border:2px solid gray;
  border-radius:15px;
  text-align:center;
  box-shadow:5px 5px 10px gray;
}

button{
  background:green;
  color:white;
  border:none;
  padding:10px 15px;
  border-radius:5px;
  cursor:pointer;
}

button:hover{
  background:red;
}

footer{
  background:black;
```

```
color:white;
text-align:center;
padding:15px;
margin-top:20px;
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<link rel='stylesheet' href='style.css'>	Links the external CSS stylesheet to the HTML document.
background: linear-gradient(to right, #0066cc, #8000ff)	Creates a two-color horizontal gradient background on the header.
nav a:hover { color: yellow; }	Changes link font color to yellow on mouse hover.
display: flex; justify-content: center; gap: 20px;	Flexbox container centering child cards with 20px uniform gap.
border-radius: 15px; box-shadow: 5px 5px 10px gray;	Rounds card corners and adds realistic drop shadows.
button:hover { background: red; }	Transitions button color from green to red upon mouse hover.

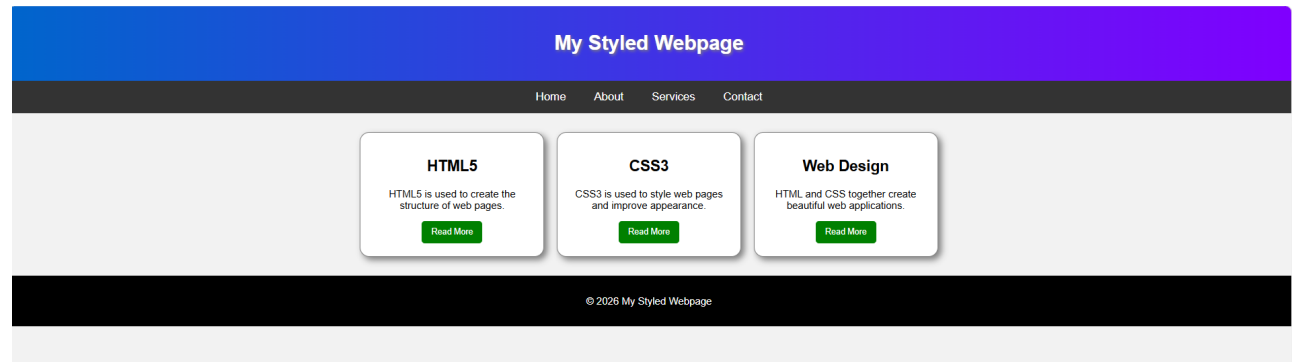
8. TEST CASES AND EXECUTION DATA

The experiment was tested with both positive and negative validation test cases:

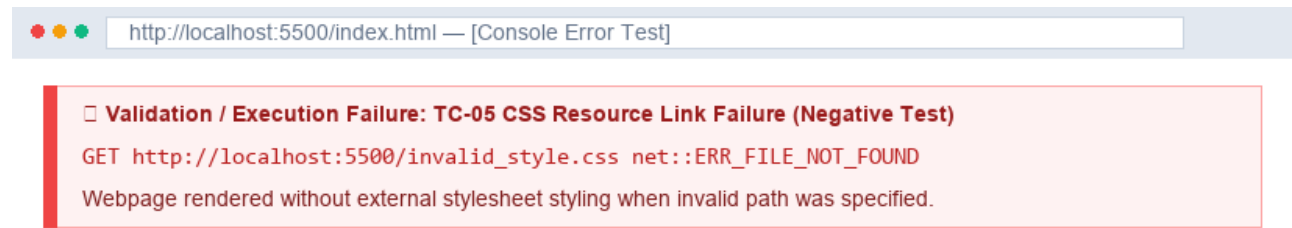
Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	External CSS Application	Load page with style.css linked	Styles, fonts, and layout rules apply uniformly to all elements	Pass
TC-02	Flexbox Card Distribution	Inspect .container flex properties	Three cards rendered side-by-side with equal 20px spacing	Pass
TC-03	Hover Transitions	Hover over navigation links and buttons	Nav link turns yellow; button background transitions green to red	Pass
TC-04	Missing CSS File	Change link href to 'missing_style.css'	Browser throws 404 ERR_FILE_NOT_FOUND and renders unstyled HTML	Fail
TC-05	Invalid Hex Color Code	Set background: '#ZZZZZZ' in CSS	CSS parser discards invalid color declaration and ignores property	Fail

9. OUTPUT

Output 1: Styled Webpage with Gradient Header, Navigation, and Flexbox Cards (Passed Output)



Output 2: Missing CSS File 404 Error / Unstyled Page Fallback (Failed Output)



10. RESULT

Thus, external CSS3 styling with linear gradients, navigation bar hover effects, and flexbox card layout was successfully implemented and verified under normal and error conditions.